4. Electromagnets

direction is not strength

A current makes a magnetic field

Lesson 4 · two 55-minute sessions · staged build; battery power only

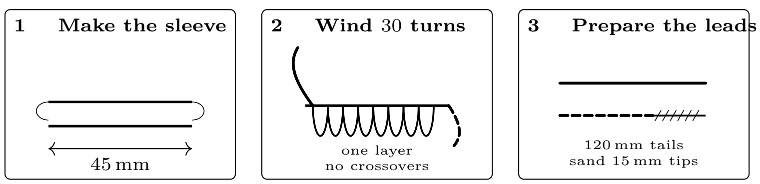
What you need

- Enamelled magnet wire, 26–30 AWG, about 4 m
- Iron bolt and a loose-fitting cardboard sleeve
- Two AA cells in a fused, switched holder
- 12 identical paper clips in one cup
- Compass and multimeter
- Sandpaper, tape, ruler, timer, index card

Predict first

A 30-turn coil lifts some paper clips. Predict what will happen when you (a) add 30 turns, (b) remove the iron core, and (c) reverse the current. For each, predict both *strength* and *pole direction*. Do not use "stronger" without naming the observation that would count as stronger.

Stage 1: build a coil that can be compared

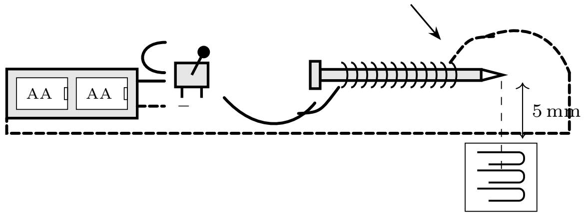


1. Roll and tape the cardboard sleeve so it slides over the bolt without wobbling. Mark a 45 mm winding zone.
2. Wind 30 adjacent turns in one direction. Tape the first and last turns. Do not wind directly on the bolt: the removable sleeve makes a fair iron-versus-air comparison possible.
3. With the battery disconnected, the adult checks that only the two lead tips are bare. Measure resistance: R_{30} = ___________ Ω. An open display means enamel or a broken wire; near zero means a bypass around the winding.

Checkpoint

Why must the winding zone, wire gauge, bolt, and test pole stay the same when turn count changes? Which of those could change resistance as well as geometry?

Stage 2: make the lifting test repeatable



index-card spacer and clip cup

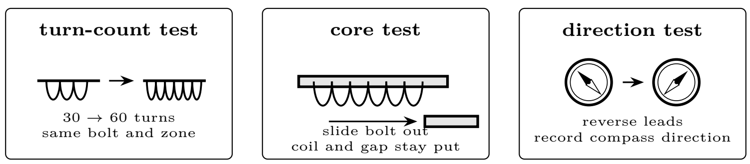
1. Put the bolt into the sleeve. Use the index card to reset the lower pole 5 mm above the same cup of clips before every run.
2. Close the switch for 3.0 s, raise the coil vertically, and count clips still attached. Open the switch immediately. Wait 30 s and return all clips before the next run.
3. Complete four baseline trials. If the spread is more than two clips, improve the method before comparing designs.

coil	current (A)	trial 1	trial 2	trial 3	trial 4	range
30 turns, iron	______	____	____	____	____	____

Pause and decide. What caused the largest uncontrolled difference between trials? Would a one-clip improvement be larger than your measurement noise?

Stage 3: separate turns, core, and direction

Add 30 turns beside the first layer, in the same winding direction and inside the same marked zone. Measure R_{60} = ________ Ω. The adult then places the ammeter in series, verifies the current jack and range, and controls the first pulse.



test	turns	core	leads	current	mean lift	range	fixed controls
A	30	iron	normal	____	____	____	pulse, gap, clips, pole
B	60	iron	normal	____	____	____	pulse, gap, clips, pole
C	60	air	normal	____	____	____	pulse, gap, clips, coil
D	60	iron	reversed	____	____	____	pulse, gap, clips, coil

Run four trials for each row. Between rows, name the one intended change aloud. For test D, hold a compass one bolt-length from the upper pole and record its direction before and after reversing the leads.

Three questions before interpreting

1. Did 60 turns also change current? Use measured NI, not turns alone, when comparing the two windings.
2. Does the iron-core difference exceed the ranges of both repeated sets?
3. Did reversing current change compass direction, lifting strength, both, or neither?

When the evidence looks wrong

symptom	inspect with power off	correction
no current, no lift	scraped tips, switch, fuse, continuity	clean only the tips; repair the open connection
high current, weak lift	wire bypass, reversed section, loose core	route current through every turn; rewind one direction
counts vary widely	gap, pulse time, mixed clips, sideways pickup	reset the card spacer, timer, cup, and vertical path
coil or cells warm	pulse duration, pinched enamel, excessive current	stop, disconnect, and have the adult inspect
compass result wanders	steel objects, distance, starting direction	clear the bench and reset the compass position

Final evidence proof

Write one conclusion that your measurements, rather than the expected theory, force you to accept.

$\boxed{\text{single planned change}} \longrightarrow \boxed{\text{difference in repeated data}} \longrightarrow \boxed{\text{difference exceeds spread}} \longrightarrow \boxed{\text{limited conclusion}}$

Planned change: ____________________ Measured difference: ____________________

Ranges compared: ____________________ Conclusion justified: ____________________

Disproof test

Name a possible next result that would make you withdraw or narrow that conclusion. A prediction confirmed once is not proof; a claim that survives a fair opportunity to fail is stronger evidence.

What the measurements mean

Current makes field Moving charge produces a magnetic field. Around one straight conductor the field circles the wire; in a coil, the fields of successive turns reinforce along the axis.

Ampere-turns matter The useful comparison is NI: turns multiplied by measured current. Added wire also adds resistance, so doubling turns does not guarantee double current-times-turns or double lifting force.

Iron changes the path Iron magnetizes strongly and concentrates flux through the bolt and the small air gap. The core test isolates that effect only because sleeve, coil, gap, clips, and pulse remain fixed.

Direction is not strength Reversing current swaps the poles. With the same current magnitude and geometry, it need not produce a large strength change. The compass and the clip count answer different questions.

The link to later coils

Relays, motors, loudspeakers, and transformers all exploit controllable fields. Later high-frequency coils use different geometry and materials, but this lesson's habits remain: measure current, control geometry, time energization, and distinguish a pole reversal from a strength change.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.